Voluntary Carbon Standard Version 2007 Verification Report Template

19 November 2007

Report No: QT-SHE030609

Verification Report:

Name of Verification company:	Date of the issue:
TÜV NORD CERT GmbH	26th Aug. 2009
Report Title:	Approved by:
Guangdong Nan Ao 26MW Wind Power Project	Mr. Eric Krupp
Client:	Project Title:
Climate Bridge Ltd.	Guangdong Nan Ao 26MW Wind Power Project
Summary:	
Climate Bridge Ltd. has commissioned the TÜV NORD JI/CDM Certification Program to carry out the verification of the project- Guangdong Nan Ao 26MW Wind Power Project, with regard to the relevant requirements of VCS 2007 Standard. The project activity is a diversion type wind project generating electricity otherwise generated by thermal power plants connected to the Southern China Power Grid (CSPG). Thus, by implementing renewable energy technologies the project activity achieves GHG emission reductions. A risk based approach has been followed to perform this verification. In the course of the verification 4 Corrective Action Requests (CAR) and 2 Clarification Requests (CL) were raised and have been successfully closed. The verification is based on the CDM PDD[/PDD/], VCS additional information[/AI/], validation report[/VVP/], monitoring report[/MR1/, /MR2/] and other supporting documents made available to the verifiers by project proponent. As a result of the verification, the verifiers confirm that: The GHG emission reduction in the monitoring period (2008-01-01 to 2008-12-14) amounting to: 48,524 tCO_{2eq}	
Work carried out by:	Number of pages:
Li Yongjun, Jiang Yijun, Lars Kirchner	16

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1 Introduction

1.1 Objective

The purpose of this verification, by independent checking of objective evidence, is as follows:

- To verify that the project is implemented as described in the project design document;
- To confirm that the monitoring system is implemented and fully function to generate voluntary carbon units (VCUs) without any double counting, and
- To establish that the data reported are accurate, complete, consistent, transparent and free of material error or omission by checking the monitoring records and the emissions reduction calculation.

1.2 Scope and Criteria

The verification of this VCS project is based on validated CDM PDD, VCS additional information, the monitoring report[/MR/], supporting documents made available to the verifier and information collected through performing interviews and during the on-site assessment. Furthermore publicly available information was considered as far as available and required.

The TÜV NORD JI/CDM CP has employed a risk-based approach in the verification, focusing on the identification of significant risks and reliability of project monitoring and generation of emission reductions.

1.3 VCS project Description

Guangdong Nan Ao 26MW Wind Power Project (hereinafter referred to as the project) is a newly built wind power station with total installed capacity of 23.25MW). The project is located in Nan Ao Island, Shantou City, Guangdong Province, People's Republic of China, and its geographical coordinates are 117°5'27.73"E and 23°25'12.63"N.

The project involves construction and operation of a wind farm with a total capacity of 23.25 MW and a 110 KV substation at Nan Ao Island. The electricity generated by the project is connected to Guangdong Provincial Power Grid and is finally transferred to China Southern Power Grid (CSPG), The net electricity exported from the project is 52,311 MWh during the monitoring period (from 1 Jan.2008 to 14 Dec.2008) with the GHG emission reduction of 48,452 tCO_2e.

1.4 Level of assurance

The verification report is based on the registered CDM PDD, VCS additional information, monitoring report, supporting documents listed as references made available to the verifier and information collected through performing interviews and during the on-site assessment. The verification opinion is assured provided the credibility of all above.

2 Methodology

Preparations: From receipt of documents to Dec 2008

On-site verification: From 2008-12-25 to 2008-12-26

Draft Reporting: 2009-08-20

Final Reporting: 2009-08-25

The verification consist the following steps:

- A desk review of the registered CDM PDD[/PDD/], VCS additional information [/AI/], the Monitoring Report [/MR/] and additional supporting documents which were submitted by the client,
- Verification audit planning,
- On-Site assessment,
- Background investigation and follow-up interviews with personnel of the project developer,
- Verification reporting (Draft Verification Report and Final Verification Report).

The criteria of this verification include the relevant rules and steps as set out in VCS 2007.1.

3 Verification Findings

3.1 Remaining issues, including any material discrepancy, from previous validation

There is no remaining issue from previous validation. The verification has been carried out based on the monitoring report [/MR/], PDD, initial verification and supporting documents provided.

3.2 Project Implementation

The project has been implemented as described in the registered PDD [/PDD/]. There are no changes in the key equipment since the validation of the project. During the monitoring period covering 2008-01-01 to 2008-12-14 the project exported 52,311MWh of net electricity to the grid, which is a part of China Southern Power Grid. This was verified by the verification team during the on site visit by checking the sale receipts of electricity, the balance sheet and the other related evidence[/EBS/]

The power generation started from Jan 2008 and the project was registered as a CDM project in 2008 Dec.15th.

Electricity generated by the project is transmitted to the local grid via the 110kV transformer substation on Nan'ao Island and then connected to CSPG.

The relevant monitoring equipment, including main and check meter are installed, state of the art and working in an appropriate manner. Additionally, relevant calibration certificates of measurement devices have been verified.

Electricity generation statements of the submitted monitoring report were prepared by consolidated monthly data over the whole monitoring period.

However, CL 1-2 were raised and successfully closed out.

Project Implementation	**CL 1**		
Classification	☐ CAR	☒ CL	☐ FAR
Description of finding *Describe the finding in unambiguous style; address the context (e.g. section)*	The description of project activity should be provided more in details, e.g. project UNFCCC reference No., registered date, project starting date, time line and evidences of turbines installation etc.		
Corrective Action #1 *This section shall be filled by the PP. It shall address the corrective action taken in details.*	More details of the project description have been added to the monitoring report.		
DOE Assessment #1 *The assessment shall encompass all open issues in annex A-1. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	The description of project activity has been detailed described in the revised monitoring report, including the UNFCCC reference number, the registered date, project starting date, timeline and evidences of turbines installation. DOE checked the relevant evidence and consulted from UNFCCC website then confirm that the revised description is accordance with real situation.		
Conclusion *Tick the appropriate checkbox*	☐ To be checked during the first periodic verification ☒ Appropriate action was taken ☐ Project documentation was corrected correspondingly ☐ Additional action should be taken ☐ The project complies with the requirements		

Project Implementation	**CL 2**		
Classification	☐ CAR	☒ CL	☐ FAR
Description of finding *Describe the finding in unambiguous style; address the context (e.g. section)*	The incorrect description of power conditioner "pitch regulated and variable speed" was detected.		
Corrective Action #1 *This section shall be filled by the PP. It shall address the corrective action taken in details.*	MR has been corrected accordingly.		
DOE Assessment #1 *The assessment shall encompass all open issues in annex A-1. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	The selected turbine model is S48/750 with pitch and speed regulated. The equipment purchasing contract has been checked. Monitoring report has been revised correctly. Therefore CL2 was successfully closed.		

Project Implementation	CL 2
Conclusion *Tick the appropriate checkbox*	☐ To be checked during the first periodic verification ☒ Appropriate action was taken ☐ Project documentation was corrected correspondingly ☐ Additional action should be taken ☐ The project complies with the requirements

3.3 Completeness of Monitoring

The parameters need to be monitored are listed as follows:

EG_{output}

Total electricity supplied to the gird during year y. The data source comes from electricity meter reading at the substation between the proposed project and the grid.

$EG_{import,y}$

Electricity purchased from the Grid by the proposed project during year *y*

EG_{output} and EG_{Import} are measured through bidirectional meter (main meter) continuously,

$EG_{new\ output,\ y}$

Total electricity generated by 8 new wind turbines during year y.

The electricity voltage is increased to 110kV first at Nan'ao project substation on Nan'ao Island, and then transmitted to the Jinniu Substation[1], which finally connects to China Southern Power Grid. Revenue meter was installed at Nan'ao project 110kV Substation as main monitoring system to measure the grid-connected power generation of the proposed project. The total net electricity supplied to the grid by the proposed project activity is the total electricity generated by the proposed project minus the electricity imported from the grid.

During the project VCU crediting period, another local wind farm company, Shenzhen Nanguo Energy Co., installed eight new turbines with each capacity of 750 kW next to the project site. The 8 new wind turbines share the 110 kv substation and the same ammeter established by the project. Therefore, since October the total net electricity supplied to the grid by the proposed project activity has been changed to be calculated by deducting the total amount of electricity generated by the eight wind turbines from the total net electricity supplied to the grid which is the difference of the electricity exported and imported from both the project activity and the 8 new turbines.

[1] In the registered PDD, it is indicated that the proposed project will connect to the Niutouling substation , However, as per onsite information and Power Purchase Agreement and Grid Connection Agreement, the project connects to Jinniu substation.The change will not impact the VCU calculation and the mode of operation .

Figure 3.1 Main connection and the allocation of metering points

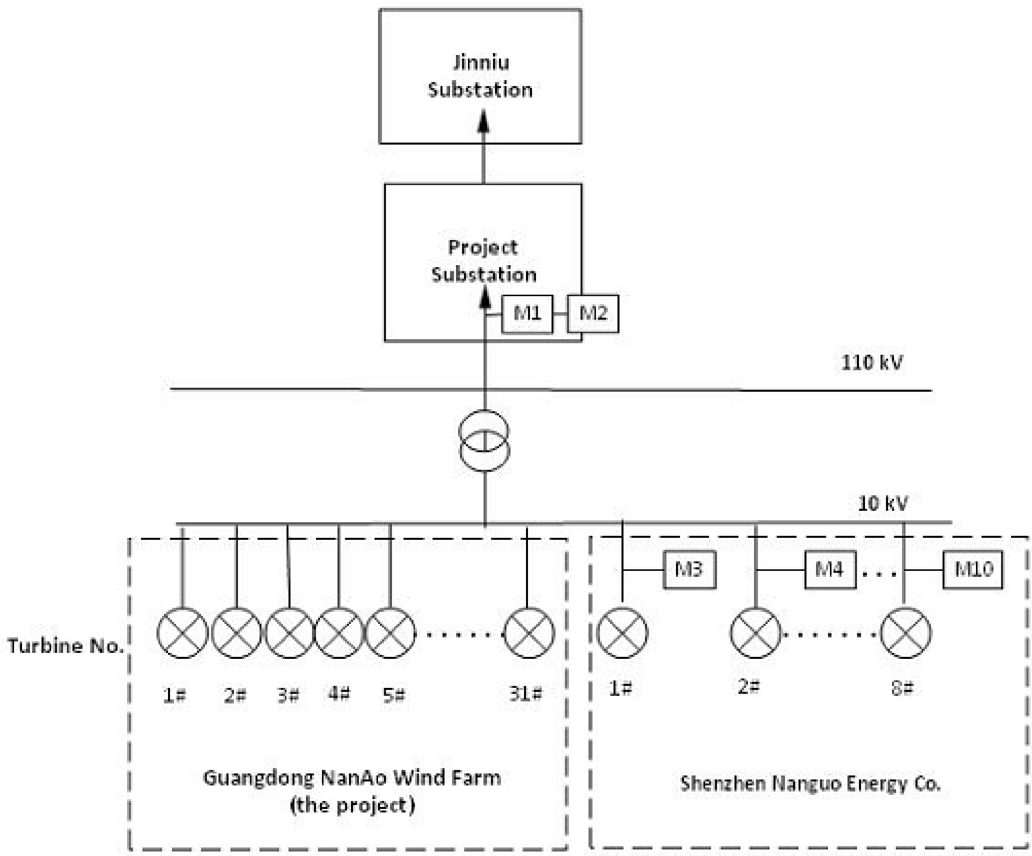

Table 3-1 Meter details

Name	Earliest Power Generation Date	Model	Accuracy
M1	2008-01-01	ZMD405	0.5S
M2(backup meter of M1)	2008-01-01	ZMD405	0.5S
M3	2008-10-09	PM810	2S
M4	2008-11-03		2S
M5	2008-10-21		2S
M6	2008-10-22		2S
M7	2008-10-27		2S
M8	2008-10-30		2S

M9	2008-10-5		2S
M10	2008-10-14		2S

The Revenue meter M1 is used as invoice meter and the meter reading is recorded on 27th or 28th of each month at 20:00. It is bi-directional and with accuracy of 0.5S. Electricity summary and invoices for the whole crediting period were available.

The revenue meter and backup meter were examined and calibrated annually according to relevant standards and regulations. The calibration was conducted by Shantou Power Company. The eight PM810 meters were installed by the manufacturer and adjusted onsite to ensure the usability. All relevant evidences have been checked by DOE.

According to the technical specification of the meter PM810 and the wind turbine, Schneider Electric's PM810 power meter meets the standard IEC61036, which equals to China's Alternating Current Static Watt-hour Meters for Active Energy (classes 1 and 2) (GB/17315-2002/IEC 61036:2000).

For conservative consideration, as the accuracy of the 8 new turbines meters is lower than the revenue meter (0.5s), the reading of 8 new turbines meters will multiply with a coefficient of (1+2%) according to the electricity standard.

In case the main meter operates abnormally, the readings from the backup meter will be used. The operation log and maintenance records have been checked by DOE. No abnormal operation has occurred during the VCU crediting period.

The revenue meter will continuously monitor the output and input of the proposed project and 8 new turbines. It is recorded by the grid company on 27th or 28th of each month at 20:00. However, the electricity generations of 8 new turbines are recorded by on-site staff every day at 20:00.Thus, the reading of 2008-12-15 20:00 will be used for the ER calculation during the crediting period 2008-01-01 to 2008-12-14 for conservativeness.

During the verification process, CAR1-4 were raised

Completeness of Monitoring	**CAR1**		
Classification	☒ CAR	☐ CL	☐ FAR
Description of finding *Describe the finding in unambiguous style; address the context (e.g. section)*	The monitoring parameters and monitoring plan should be revised due to the fact of joint meter reading.		
Corrective Action #1 *This section shall be filled by the PP. It shall address the corrective action taken in details.*	Monitoring parameters have been revised in the monitoring report according to the joint meter reading.		

Completeness of Monitoring	**CAR1**
DOE Assessment #1 *The assessment shall encompass all open issues in annex A-1. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	The monitoring parameters and monitoring plan have been revised according to the real situation. **EG** $_{\text{new output, y}}$ has been added as a monitoring parameter. The electricity generation from Shenzhen Nanguo Energy Co.'s 8 new turbines is monitored separately by the meter installed at each wind turbine. The total electricity supplied to the grid by the project activity and the 8 new turbines which started operation since Oct. 2008 and are owned by Shenzhen Nanguo Energy Co. is monitored by the main meter installed at the substation. Thus, from January through September 2008, the total net electricity supplied to the grid by the proposed project activity is the total electricity generated by the proposed project minus the electricity imported from the grid which were both monitored by the ammeter at the substation. And during the period 2008-10-05 to 2008-12-14 the total net electricity supplied to the grid by the proposed project activity is calculated by deducting the total amount of electricity generated by the eight wind turbines from the total net electricity supplied to the grid which is the difference of the electricity exported and imported from both the project activity and the 8 new turbines As the project will take the shared line losses of the Shenzhen Nanguo Energy Company, the conservativeness of the emission reduction calculation has been guaranteed. Therefore, CAR1 was successfully closed out.
Conclusion *Tick the appropriate checkbox*	☐ To be checked during the first periodic verification ☒ Appropriate action was taken ☒ Project documentation was corrected correspondingly ☐ Additional action should be taken ☐ The project complies with the requirements

Completeness of Monitoring	**CAR2**		
Classification	☒ CAR	☐ CL	☐ FAR
Description of finding *Describe the finding in unambiguous style; address the context (e.g. section)*	1) The meters calibration records covering the monitoring period should be submitted. 2) The qualification of the calibration entity should be proven. 3) Meters diagram should be provided.		
Corrective Action #1 *This section shall be filled by the PP. It shall address the corrective action taken in details.*	1) the calibration records have been submitted to the auditor 2) the document on calibration entity qualification has been submitted to the auditor 3) meter diagram has been provided to the auditor.		

Completeness of Monitoring	CAR2
DOE Assessment #1 *The assessment shall encompass all open issues in annex A-1. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	The meters calibration records of revenue meter and backup meter covering the monitoring period have been submitted to DOE. DOE also checked the debugging reports of 8 new turbines to ensure usability and correctness of the PM810 meter inside each wind turbine. The qualification of calibration entity and the calibrator's certification have been submitted. The meter diagram has been added into revised MR Therefore, CAR2 is successfully closed
Conclusion *Tick the appropriate checkbox*	☐ To be checked during the first periodic verification ☒ Appropriate action was taken ☒ Project documentation was corrected correspondingly ☐ Additional action should be taken ☐ The project complies with the requirements

Completeness of Monitoring	CAR3		
Classification	☒ CAR	☐ CL	☐ FAR
Description of finding *Describe the finding in unambiguous style; address the context (e.g. section)*	The emission reduction calculation, monitoring period and pre-registration PD should be revised as per CDM registration.		
Corrective Action #1 *This section shall be filled by the PP. It shall address the corrective action taken in details.*	Relevant calculation in MR has been revised when the project got CDM registration.		
DOE Assessment #1 *The assessment shall encompass all open issues in annex A-1. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	The monitoring period and pre-registration have been revised. Concerning the emission reduction calculation, the verifiers checked the balance sheet and the monitoring records of 8 new turbines of Nan'guo Company and cross checked with sales invoices and confirmation notice both provided by the grid company. Hereby the verifiers confirm that the ER calculation is correct and conservative.		
Conclusion *Tick the appropriate checkbox*	☐ To be checked during the first periodic verification ☒ Appropriate action was taken ☒ Project documentation was corrected correspondingly ☐ Additional action should be taken ☐ The project complies with the requirements		

Completeness of Monitoring	CAR4		
Classification	☒ CAR	☐ CL	☐ FAR
Description of finding *Describe the finding in unambiguous style; address the context (e.g. section)*	More information on project and measurement QA/QC procedures should be provided in MR.		

Completeness of Monitoring	CAR4
Corrective Action #1 *This section shall be filled by the PP. It shall address the corrective action taken in details.*	More detailed QA/QC procedure has been added to the monitoring report.
DOE Assessment #1 *The assessment shall encompass all open issues in annex A-1. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	QA/QC procedure have been detailed addressed into revise monitoring report The total output and input electricity generations are continuously monitored by the main meter and recorded on 27th or 28th of each month at 20:00 by the grid company. It can be cross checked with sales invoices and balance sheet. The electricity generations of 8 new turbines are recorded by on-site staff every day at 20:00. The terminal computer system also records the power generation from each turbine of the project activity and of 8 new turbines. It could be cross check with the revenue meter readings for electricity exported minus electricity imported then deduct the electricity generated by the proposed project which is evidenced by the grid company. The revenue meter and backup meter were examined and calibrated annually according to relevant standards and regulations. The calibration was conducted by Shantou Power Company. The eight PM810 meters were installed by the manufacturer and adjusted onsite to ensure the usability In case the main meter operates abnormally, the readings from the backup meter will be used. All data and records of meter readings, meter calibration, and other documentation will be archived in both electronic and written forms, and be kept by the project owner for 2 years after the end of the last crediting period.
Conclusion *Tick the appropriate checkbox*	☐ To be checked during the first periodic verification ☒ Appropriate action was taken ☒ Project documentation was corrected correspondingly ☐ Additional action should be taken ☐ The project complies with the requirements

3.4 Accuracy of Emission Reduction Calculations

According to the validated CDM PDD, the approved CDM baseline and monitoring methodology ACM0002 (Version 06), is applied to the project.

GHG emission reduction is calculated as baseline emissions minus project and leakage emissions. Both, the project and leakage emissions can be considered as 0, in compliance with ACM0002.

Baseline emissions are achieved by multiplying net electricity delivered to the grid by the emission factor.

The EF is calculated in the registered PDD and the EF under the CDM is valid for the relevant monitoring period under the VCS, too and poses 0.9276 (tCO_2e/MWh).

It was verified in the course of this verification that the above mentioned methodology has been correctly and accurately applied in calculating the total emission reductions.

3.5 Quality of Evidence to Determine Emission Reductions

The revenue meter readings are carried out on 27th or 28th of each month at 20:00. All relevant evidences were fully checked by the verification team during the on-site visit. All evidences are clearly identifiable and assessed to be correct.

All records needed for monitoring are archived in line with the requirements of the registered monitoring plan. It could be evidenced that the monitoring system ensures for continuous (except for some routine breakdowns or outage) operation.

3.6 Management and Operational System

The allocation of responsibilities is documented in monitoring manual. Routines for the archiving of data are defined and documented. CDM manager, Mr Lin Weiping is responsible for the implementation of the monitoring plan.

Staff training plan is included in the monitoring manual, training records have been checked by DOE.

4 Verification conclusion

Climate Bridge Ltd. has commissioned the TÜV NORD JI/CDM Certification Program to carry out the verification of the project- Guangdong Nan Ao 26MW Wind Power Project with regard to the relevant requirements of VCS 2007 Standard. The scope of this verification covers the determination of GHG emission reductions generated by the project.

In the course of the verification 4 Corrective Action Requests (CAR) and 2 Clarification Requests (CR) were raised and successfully closed.

As a result of the verification, the verifier confirms that:

All operations of the project are implemented and installed as planned and functional. The installed equipment essential for generating emission reductions runs reliable is calibrated appropriately.

The GHG emission reductions are calculated without material misstatements in a conservative and appropriate manner.

Reporting period: From 2008-01-01 to 2008-12-14

Verified emission in the above reporting period:

Project emissions	0	t CO_2 equivalents
Baseline emissions	48, 524	t CO_2 equivalents
Emission reductions	48,524	t CO_2 equivalents

5 References

Table 5-1: Project related documents

Reference	Document
/AEIA/	Application of Environment Impact Assessment carried out by Shantou Environment Protection Institute on 21/07/2004 and approved by Shantou Environment Protection Bureau on 13/08/2004
/AEP/	Approval of Expansion Project in Dannan Wind Power Plant issued by Plan and Soil Resource Bureau in Nan'ao County dated on 16/09/2008 Ref :Nan Gui Guo Yong [2008]35
/AL/	Additional information of the project regarding to VCS criteria
/ASTL/	Agreement of sharing transmission losses and the imported electricity of 8 new turbines from the grid , signed between Shenzhen Nanguo Power Company and China Resources Wind Power□Shantou□ Co., Ltd
/BL/	Business License of China Resources Wind Power Development Co., Ltd.
/EBS/	-Electricity Balance Sheet provided by grid company for the period Jan 2008 to Dec. 2008 -Electricity Balance Sheet provided by grid company for the period 2008-11-28 to 2008-12-14 -Sale Receipts of Electricity -Confirmation of electricity generated of the project by Grid Company for the period 2008-09-27 20:00 to 2008-12-14 24:00 -Operation manual records for 8 new turbines of Nanguo -Computer records for 8 new turbines of Nanguo
/EPC/	Equipment Purchase Contract Technical Specification Contract Technical Specification Contract of PM810
/LOG/	Sample Copy of Project Operation Records
/MCR/	Maintenance & Calibration Records -voltage transformer and current transformer calibration record -the qualification of the calibration entity -debugging report of the 8 new turbines
/MM/	Monitoring Manual
/MR01/	Monitoring Report.(Dec 18th 2008)
/MR02/	Monitoring Report.(Aug 26th 2009)
/PCD/	Power Connection Diagram

Reference	Document
/**PDD**/	Registered Project Design Document of Guangdong Nan Ao 26MW Wind Power Project
/**PPA**/	Power Purchasing Agreement (2008)
/**RWCP**/	Reply of Water-Soil Conservation Plan for Expansion project in Dannan Wind Power Plant issued by Water Conservancy Bureau in Nan'ao County dated on 24/09/2008
/**OCCA**/	Opinion of Construction Content Adjustment in Dannan Wind Power Plant issued by Shantou Environment Protection Bureau dated on 18/11/2008 Ref Shan Shi Huan Han[2008]413
/**SAR**/	Staff Assignment & Responsibilities
/**TALT**/	Tenancy Agreement of Lines and Transformer Station between Shenzhen Nanguo Power Company and China Resources Wind Power□Shantou□ Co., Ltd
/**TCR**/	Staff Training and Competence Records (2008) 1. Staff Training Plan 2. Staff Training Records 3. Sample Copy of Operator Certificate
/**XLS**/	The Emission Reduction Calculation Spreadsheet

Table 5-2: Background investigation and assessment documents

Reference	Document
/**ACM**/	Approved CDM Methodology ACM0002(Version06)
/**VAL**/	Validation Report for CDM project 'Guangdong Nan Ao 26MW Wind Power Project'
/**VVM**/	Validation and Verification Manual
/**PDD**/	Validated Project design document for CDM project: Guangdong Nan Ao 26MW Wind Power Project'

Table 5-3: Websites used

Reference	Link	Organisation

Reference	Link	Organisation
/**unfccc**/	http://cdm.unfccc.int	UNFCCC
/**vcs**/	www.v-c-s.org	VCS website

Table5- 4: Interviewed persons

Reference		Name	Organisation / Function
/**IM01**/	☒ Mr. ☐ Ms.	Mr. Lin Weiping	China Resources Wind Power Development Co., Ltd. Vice Director of CDM Center
/**IM02**/	☐ Mr. ☒ Ms.	Ms. Li Ling	Project Manager, Climate Bridge Ltd.